Sarcasm Detection in Social Media using Multi-Task Learning

Project Output Screenshots · Read-Only Reference

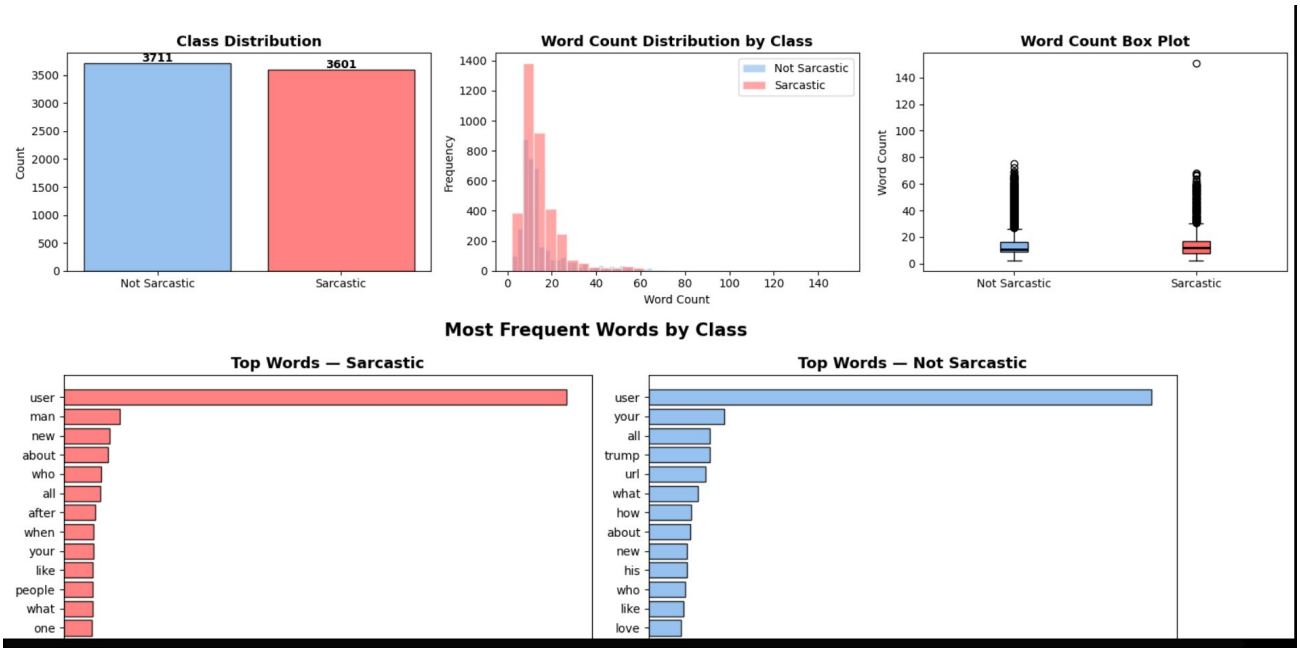
BERT + Sentiment Auxiliary Task · Google Colab

Model Performance Summary

Model	Accuracy	Precision	Recall	F1-Score
Baseline 1: TF-IDF + Logistic Regression	0.7712	0.7613	0.7796	0.7704
Baseline 2: BERT (single-task)	0.8505	0.8500	0.8500	0.8453
Proposed: BERT MTL + Sentiment Feature	**0.8259**	**0.8226**	**0.8241**	**0.8233**

Output 1 — Exploratory Data Analysis (Cell 4)

Class distribution, word count histogram, box plot, and top frequent words per class.



Output 2 — Baseline 1: TF-IDF + Logistic Regression (Cell 10)

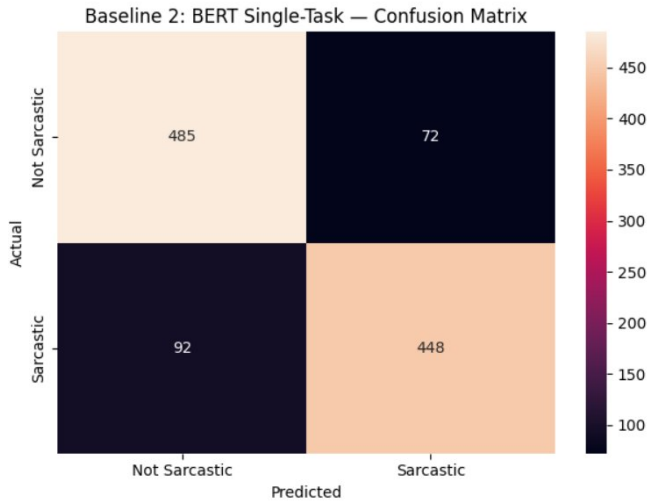
Classification report showing accuracy, precision, recall and F1 for the TF-IDF baseline.

```
=========================================================
BASELINE 1: TF-IDF + Logistic Regression
=========================================================
Accuracy : 0.7712
Precision: 0.7613
Recall   : 0.7796
F1 Score : 0.7704
              precision    recall  f1-score   support

Not Sarcastic       0.78      0.76      0.77       557
    Sarcastic       0.76      0.78      0.77       540

     accuracy                           0.77      1097
    macro avg       0.77      0.77      0.77      1097
 weighted avg       0.77      0.77      0.77      1097
```

Output 3 — Baseline 2: BERT Single-Task (Cell 11)

Classification report and confusion matrix for the BERT single-task model.

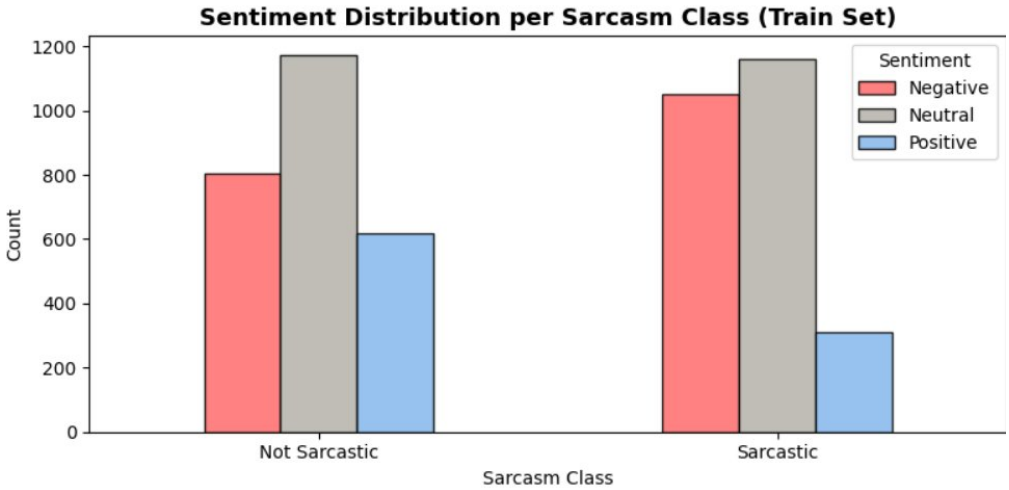
```
Accuracy : 0.8505
F1 Score : 0.8453
              precision    recall  f1-score   support

Not Sarcastic       0.84      0.87      0.86       557
    Sarcastic       0.86      0.83      0.85       540

     accuracy                           0.85      1097
    macro avg       0.85      0.85      0.85      1097
 weighted avg       0.85      0.85      0.85      1097
```



Output 4 — Sentiment Distribution per Sarcasm Class (Cell 12)

RoBERTa sentiment labels distributed across sarcastic and non-sarcastic samples in the training set.

```
Sentiment label distribution (train):
  Negative  : 1855
  Neutral   : 2334
  Positive  : 929
```

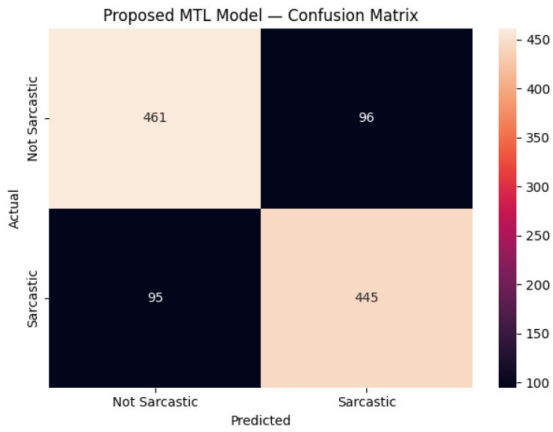


Output 5 — Proposed BERT MTL Model (Cell 15)

Classification report and confusion matrix for the proposed multi-task learning model.

```
===== MTL Model — Test Set Evaluation =====
/tmp/ipykernel_565/3905935217.py:13: FutureWarning: `torch.cuda.amp.
  with torch.cuda.amp.autocast(enabled=use_fp16):
Accuracy : 0.8259
Precision: 0.8226
Recall   : 0.8241
F1 Score : 0.8233
              precision    recall  f1-score   support

Not Sarcastic       0.83      0.83      0.83       557
    Sarcastic       0.82      0.82      0.82       540

     accuracy                           0.83      1097
    macro avg       0.83      0.83      0.83      1097
 weighted avg       0.83      0.83      0.83      1097
```

colab.res

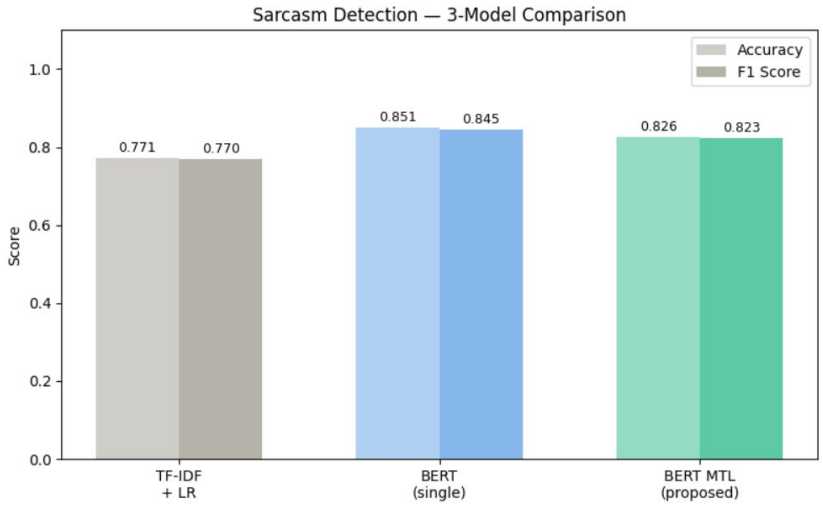


Output 6 — Final 3-Model Comparison Summary (Cell 16)

Side-by-side accuracy and F1-score comparison across all three models.

```
============================================================
              FINAL MODEL COMPARISON SUMMARY
============================================================
Model                                      Accuracy       F1
------------------------------------------------------------
Baseline 1: TF-IDF + Logistic Regression     0.7712   0.7704
Baseline 2: BERT (single-task)               0.8505   0.8453
Proposed:   BERT MTL + sentiment feature     0.8259   0.8233
============================================================

MTL did not improve over BERT (2.20% gap). Try tuning ALPHA.
```

colab.research.google.com – To exi



Output 7 — Inference Demo: 9/9 Correct (Cell 18)

Live predictions using the BERT MTL + rule-based ensemble. All 9 test cases correctly classified.

```
==========================================================================================
✓  [Sarcastic     ]  I absolutely love being stuck in traffic for two hours.
        -> Sarcastic (bert: 96.7% | rules: 60.0% | ensemble: 96.7% | sentiment: Positive)

✓  [Sarcastic     ]  Scientists discover water is wet.
        -> Sarcastic (bert: 92.9% | rules: 80.0% | ensemble: 92.9% | sentiment: Neutral)

✓  [Sarcastic     ]  Sure, that plan is totally going to work out perfectly.
        -> Sarcastic (bert: 98.1% | rules: 60.0% | ensemble: 98.1% | sentiment: Positive)

✓  [Sarcastic     ]  Oh great, another Monday morning meeting.
        -> Sarcastic (bert: 97.5% | rules: 72.0% | ensemble: 97.5% | sentiment: Positive)

✓  [Not Sarcastic ]  The weather is nice today.
        -> Not Sarcastic (bert: 5.3% | rules: 0.0% | ensemble: 5.3% | sentiment: Positive)

You seem to be using the pipelines sequentially on GPU. In order to maximize efficiency please use a dataset
✓  [Not Sarcastic ]  The food at the restaurant was delicious.
        -> Not Sarcastic (bert: 3.6% | rules: 0.0% | ensemble: 3.6% | sentiment: Positive)

✓  [Sarcastic     ]  Wow, what a surprise, the train is late again.
        -> Sarcastic (bert: 98.1% | rules: 72.0% | ensemble: 98.1% | sentiment: Negative)

✓  [Sarcastic     ]  I just love working overtime every single weekend.
        -> Sarcastic (bert: 93.2% | rules: 60.0% | ensemble: 93.2% | sentiment: Positive)

✓  [Not Sarcastic ]  The team played really well today and won.
        -> Not Sarcastic (bert: 4.9% | rules: 0.0% | ensemble: 4.9% | sentiment: Positive)

Demo accuracy: 9/9 (100%)
```